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# Research Statement

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## 1. BLACK HOLE MASS SCALING RELATIONS

### 1.1. *The $M_{\rm BH}$–$\phi$ Relation*

Seigar et al. (2008) first presented evidence of a relationship between the mass of a supermassive black hole, ($M_{\rm BH}$) and the pitch angle ($\phi$) of its host galaxy. This introduced us to a powerful tool to estimate the masses of black holes in distant galaxies using only simple imaging data, without the need for telescope-expensive spectral data. In Berrier et al. (2013), we increased the number of spiral galaxies with directly measured supermassive black holes (SMBHs) from 5 (as it was in Seigar et al. 2008), to 22. In Davis et al. (2014), we generated a black hole mass function (BHMF) for local spiral galaxies by applying the updated $M_{\rm BH}$–$\phi$ relation to a volume-limited sample of 144 galaxies from the Carnegie-Irvine Galaxy Survey (Ho et al. 2011). In a follow-up study (Mutlu-Pakdil et al. 2016), we also derived the BHMF for the complementary sample of 68 early-type galaxies. In Davis et al. (2017), we doubled the number of directly measured black holes used to generate the $M_{\rm BH}$–$\phi$ relation.[1] This work received favorable coverage in the media, appearing in no less than 44 outlets,[2] including one magazine article I was requested to write for *Australasian Science*.[3]

### 1.2. *The $M_{\rm BH}$–$M_{\star,\rm sph}$ Relation*

My main attraction to moving to Australia was the opportunity to work with Alister Graham. I consider him the world's foremost expert when it comes to multicomponent decomposition of galactic light profiles and the modeling of their spheroids with Sérsic profiles. I have learned much from him in my three years here at Swinburne; I have become adept at the aforementioned skills and the cutting-edge software (Ciambur 2015, 2016) that our research group utilizes. The majority of studies, particularly large surveys and pipelines, perform simple, automated, two-component (i.e., bulge/disk) decompositions to galaxies. Our research shows that this näiveté and oversimplification can be harmfully misleading and is often wrong.

I recently completed my magnum opus by performing meticulous multicomponent decompositions of the spiral galaxy sample from Davis et al. (2017). In Davis et al. (2019a), we presented these detailed decompositions and our unprecedentedly accurate stellar bulge masses ($M_{\star,\rm sph}$). From our resulting sample, we performed a custom, sophisticated Bayesian linear regression to define the $M_{\rm BH}$–$M_{\star,\rm sph}$ relation for spiral galaxies, such that $M_{\rm BH} \propto M_{\star,\rm sph}^{2.44\pm0.33}$ (left panel of Figure 1). This slope is significant due to its marked difference from the slope we found for early-type galaxies, $M_{\rm BH} \propto M_{\star,\rm sph}^{1.27\pm0.07}$ (Sahu et al. 2019). Since the latter relation is almost ubiquitously reproduced in large $N$-body galaxy simulations for *all* types of galaxies, our research has important implications for such simulations; simulations that tune their results to a universal linear rate of growth between bulge masses and black holes need revision.

### 1.3. *The $M_{\rm BH}$–$M_{\star,\rm tot}$ and $M_{\rm BH}$–$M_{\star,\rm disk}$ Relations*

We released our study of bulge stellar masses in tandem with our study of total ($M_{\star,\rm tot}$) and disk ($M_{\star,\rm disk}$) stellar masses Davis et al. (2018), where we presented evidence that the $M_{\rm BH}$–$M_{\star,\rm tot}$ relation is steeper, but not significantly less accurate than the $M_{\rm BH}$–$M_{\star,\rm sph}$ relation for spiral galaxies (middle panel of Figure 1). This is advantageous because it implies that black holes can be reliably predicted by measuring $M_{\star,\rm tot}$, a task that is significantly easier than isolating the stellar mass of its bulge. This can be of particular use to large surveys and pipelines, for whom, producing thousands of time-consuming decompositions would be prohibitively time-intensive. We also presented evidence that $M_{\rm BH}$ does weakly correlate with $M_{\star,\rm disk}$ (right panel of Figure 1), a relation which was previously thought to not exist (e.g., Kormendy & Gebhardt 2001).

### 1.4. *The $M_{\rm BH}$–$v_{\rm max}$ and $M_{\rm BH}$–$M_{\rm DM}$ Relations*

I have previous experience working with dark matter halo masses from my work five years ago (Seigar et al. 2014). Some of my current research involves investigating the connection between spiral galaxy structure,

[1] See the following link for a schematic of the $M_{\rm BH}$–$\phi$ relation, https://bendavis007.github.io/images/Schematic.pdf.

[2] https://bendavis007.github.io/news.html

[3] https://www.australasianscience.com.au/article/issue-sepoct-2017/how-you-can-weigh-black-holes.html

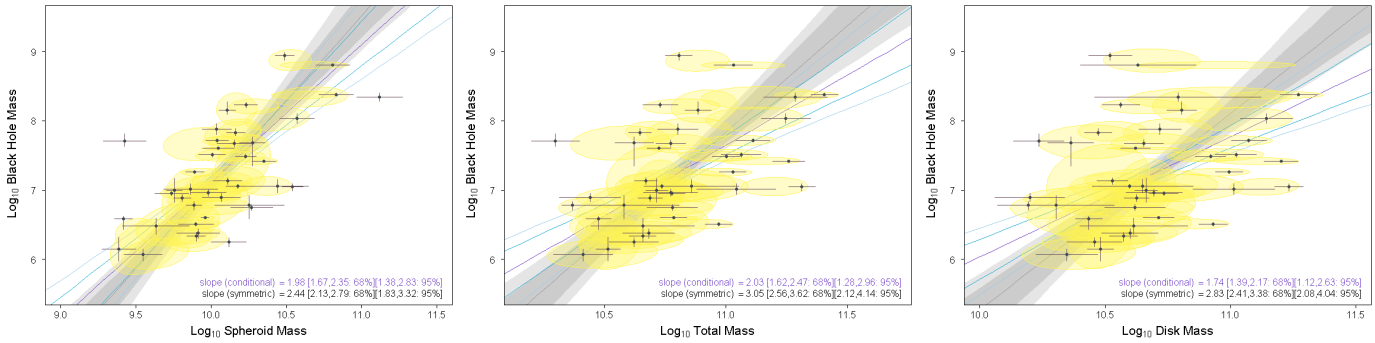


**Figure 1. Left** – The *symmetric* line of best fit for the $M_{\rm BH}$–$M_{\star,\rm sph}$ relation is presented as its pointwise median with ±68% and ±95% (shaded) intervals, while the ±68% posterior estimates of the true stellar spheroid and black hole mass of each galaxy are highlighted in yellow. The *conditional* (purple) line of best fit is additionally supplied with similar (cyan) error intervals. Masses are in units of solar masses. **Middle** – Here, the $M_{\rm BH}$–$M_{\star,\rm tot}$ relation is presented. **Right** – Here, the $M_{\rm BH}$–$M_{\star,\rm tot}$ relation is presented.

black hole mass, and dark matter halo mass ($M_{\rm DM}$). Kennicutt (1981) first presented evidence that $\phi$ correlates with the maximum rotational velocity ($v_{\rm max}$) of disk galaxies. Ferrarese (2002) initially showed that black hole mass correlates with $M_{\rm DM}$. In my most recent work (Davis et al. 2019b), I have updated both of these relations (see Figure 2).

### 1.5. *Fundamental Planes*

Spiral density wave theory has existed, and been the most popular theory governing spiral structure in disk galaxies, for more than a half-century now. Lin & Shu (1966) made a specific prediction that $\phi$ should be dictated by both the bulge mass and density of gas in the disk of a spiral galaxy. In Davis et al. (2015), we observationally confirmed this prediction.[4] Thus, the spiral density wave depends on the density of the disk, rather than the total mass of the disk, and it is the mass of the bulge that effectively anchors the spiral arm, a bit like setting the tension in the vibrating string of a violin by adjusting the tuning peg. Armed with increased data, my research group will be actively exploring trivariate relations between galactic parameters (e.g., van den Bosch 2016).

### 1.6. *Searching for Intermediate-mass Black Holes*

Recently, I was involved with an international collaboration to use multiple scaling relations, including X-ray and radio luminosity relations, to search for possible intermediate-mass black holes, IMBHs (Koliopanos et al. 2017). Follow-up research has included the *Chandra X-ray Observatory's* Cycle 18 Large Program titled "Spiral galaxies of the Virgo cluster," which will image 52 galaxies with the ACIS-S detector. In Graham et al. (2018), we predicted the central black hole masses in these galaxies using the latest black hole scaling relations involving $\phi$, velocity dispersion, and $M_{\star,\rm tot}$; focused study of strong candidates is ongoing. We have bolstered this sample by using suitably deep archival *Chandra* images for an additional 22 spiral galaxies in the Virgo cluster, providing a complete sample of 74 spiral galaxies with star-formation rates $> 0.3\,M_\odot\,{\rm yr}^{-1}$. Ongoing research is also being conducted in collaboration with Bogdan Ciambur (Observatoire de Paris), involving a focused study of 40–50 IMBH-candidate galaxies from (Graham & Scott 2013) and the bulgeless LEDA 87300 (Graham et al. 2016). Future work will include measuring pitch angles for all of the later type spiral galaxies (i.e., Hubble type Sd) in the Third Reference Catalogue of Bright Galaxies (de Vaucouleurs et al. 1991).

### 1.7. *Evolution of Spiral Galaxies*

Future work will focus on extending the local $M_{\rm BH}$–$\phi$ relation to high redshifts. One of my colleagues, Tiantian Yuan, is an expert at source-plane reconstruction of gravitationally lensed high-redshift spiral galaxies. She recently discovered the most distant spiral galaxy to date, at $z = 2.54$ (Yuan et al. 2017). Tiantian's discoveries have opened up a new world of discovery for spiral galaxies. Just a few years ago, spiral galaxies were thought not to exist beyond $z = 2$, and the spirals that were found from that epoch were thought not to be "true" spirals, just conglomerations of clumpy gas clouds (e.g., Elmegreen & Elmegreen 2014). Tiantian has compiled a list of 20+ spiral galaxies in the redshift bins of $z = 0.6$–$1.2$ and $z = 1.2$–$1.8$, plus a few at $z = 1.8$–$2.5$. She has adaptive-optics-assisted, Keck integral field unit spectral data for some of these already, and hence rotational information. I am measuring pitch angles for these galaxies, and we are also artificially red-

[4] See the following link for an animated gif of the 3D relation, https://bendavis007.github.io/images/plane.gif.

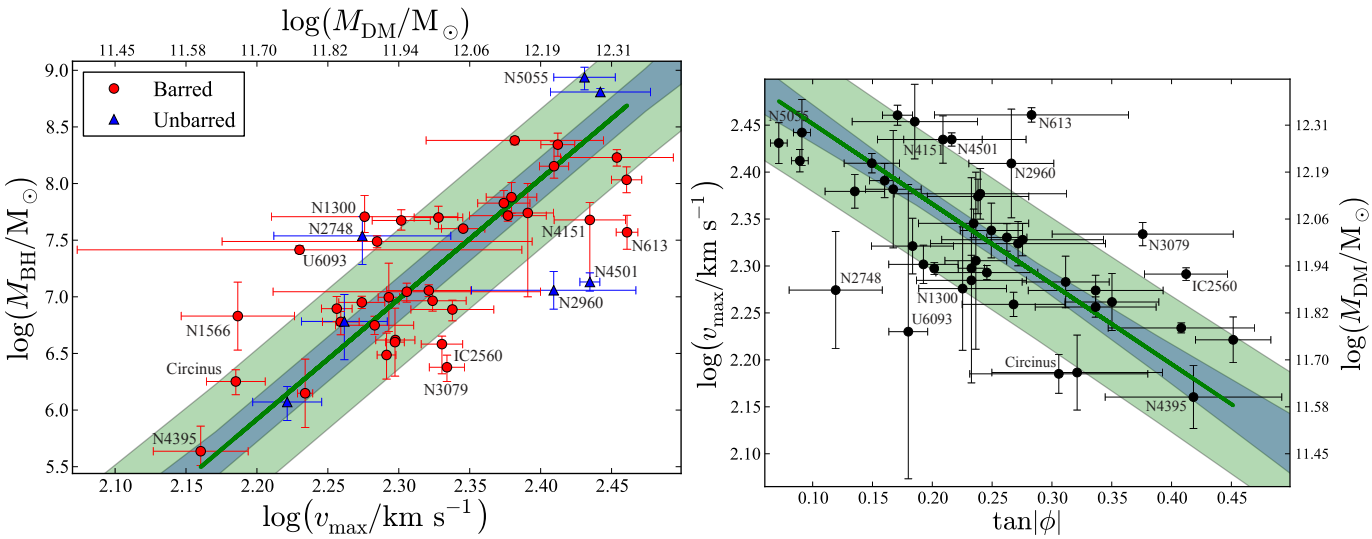


**Figure 2. Left** – Black hole mass vs. maximum rotational velocity (and dark matter halo mass) for 42 galaxies. Galaxies with barred morphologies are represented by ● and those without bars are represented by ▲. **Right** – Maximum rotational velocity (and dark matter halo mass) vs. the tangent function of the spiral-arm pitch angle absolute value for our sample of 42 spiral galaxies.

shifting local spiral galaxies to similar redshifts as the high-redshift sample, from which we will draw comparisons. We intend to construct a high-$z$ $\phi$–$v_{\rm max}$ relation, and $\phi$–$M_{\star,\rm tot}$ relation, and compare it with the local $z \sim 0$ relations that we already have. We have initiated a collaboration with Elena D'Onghia to complement our research with her input on the interpretation, potential evolutionary pathways, and additional ideas/science.

## 2. SPIRAL ARM THEORY TESTING

Spiral density wave theory predicts that when measured in different wavelengths of light, spiral arms will display slightly different pitch angles. In Pour-Imani et al. (2016); Miller et al. (2019), we measured the pitch angles for spiral galaxies in multiple passbands from the far-ultraviolet to 8.0 $\mu$m. We found that $\phi$ was statistically tightest in 3.6 $\mu$m, and loosest in 8.0 $\mu$m images. I recently started a collaboration with the Physics at High Angular resolution in Nearby GalaxieS (PHANGS)[5] research group to build upon the findings of Pour-Imani et al. (2016); Miller et al. (2019). I am working with Tiantian Yuan, Guillermo Blanc, Eric Emsellem, Sharon Meidt, Eva Schinnerer, I-Ting Ho, Kathryn Kreckel, and Miguel Querejeta; I am leading an effort to directly measure spiral density waves observed in multiple wavelengths of electromagnetic radiation.

With the PHANGS group, I will utilize their high-resolution carbon monoxide imaging from the Atacama Large Millimeter/submillimeter Array (ALMA) and H$\alpha$ imaging from the Multi-Unit Spectroscopic Explorer (MUSE), installed at the Very Large Telescope (VLT), to measure the location of the spiral density wave in face-on, grand design spiral galaxies, with unparalleled precision. Recently, Peterken et al. (2019) presented evidence that pattern speed does not vary with radius, proving that a spiral density wave governs it. With our superior data with PHANGS, combined with pitch angle information, we should be able to surpass the quality of their test of spiral density wave theory. In addition to this observational test of spiral density wave theory, I am simultaneously collaborating with Fiorenzo Vincenzo, an expert in the chemical, dynamical, and spectro-photometric evolution of galaxies within cosmological hydrodynamical simulations, to independently test spiral density wave theory via simulations.

### 2.1. *Logarithmic Spiral Arm Pitch Angle*

Logarithmic spiral arm pitch angle was the focus of my Ph.D. thesis (Davis 2015) and continues to be my passion and the subject for which I possess the highest level of expertise. Logarithmic spirals are ubiquitous throughout nature, manifesting themselves as optimum rates of radial growth for azimuthal winding in numerous structures such as mollusk shells, tropical cyclones[6], and the arms of spiral galaxies. See §2 from Davis et al. (2017) for a mathematical primer on $\phi$. I have been personally involved in the development of three inde-

[5] https://sites.google.com/view/phangs/home

[6] I have also measured the pitch angle of hurricanes, which turns out to correlate with the low-pressure in the eye of the storm.

pendent software packages to analyze images of galaxies and quantify their pitch angles:

1. The two-dimensional fast Fourier transform software, 2DFFT[7] (Davis et al. 2012, 2016), operates by decomposing galaxy images into logarithmic spirals and determining, for each number of harmonic modes ($1 \leq m \leq 6$), the pitch angle that maximizes the Fourier amplitude.

2. The spiral template fitting software, SPIRALITY (Shields et al. 2015a,b), operates by examining pixel values on an image along logarithmic spiral coordinate axes with an origin at the center of a galaxy. It then computes the median pixel value along every axis. Spiral axes with $\phi$ that are drastically different from the real $\phi$ of a galaxy (or with the wrong chirality) will have little or no variance amongst different axes across the phase angle space of the spiral coordinate system. However, spiral axes that closely resemble the true $\phi$ of a galaxy will have increased variance as some axes will lie upon bright spiral arms and others will lie upon the darker gaps in between spiral arms. The actual pitch angle of the galaxy will be the set of axes with a pitch angle that maximizes the variance. This software is also quite useful for counting the number of spiral arms.

3. The computer vision software, SPARCFIRE[8] (Davis & Hayes 2014), uses machine learning to identify the clusters of pixels that form the spirals in galaxy images.

## 3. FUTURE PROJECTS AND PLANS

My research is ideally suited for graduate and undergraduate students. During my time as a graduate student, I mentored an REU student each summer, whom I taught my pitch angle software and how to measure spiral galaxies. Many of these summer projects led to these students co-authoring papers with me, including Matt Hartley, Luke Johns, and Russell Flatman. As a postdoc, I am a co-supervisor for Ph.D. students: Nandini Sahu and Suei Hon; I have trained them to use our decomposition software and am actively reviewing their projects and editing their manuscripts. Many of the students I have worked with have limited experience with astronomy and possess more general scientific backgrounds. With a little work, I find it has been straightforward to train them in the methodologies necessary for my research and develop their abilities and knowledge as astronomers.

If I am indeed privileged and fortunate enough to become a professor, I aim to continue my impactful research and maintain my international collaborations, which extend to five continents now. I desire to establish a vibrant and productive extragalactic research group with passionate and motivated students. I have learned from experience that students are very interested in what is new and disruptive in science. Perhaps the most exciting new field in science today is the study of gravitational waves. I am an affiliate of OzGrav[9] and Nobel Laureate Kip Thorne is my "academic grandfather," as a result, I have a lot of connections in the field of gravitational radiation and the LIGO collaboration. As ground-based detectors become more sensitive to longer wavelength gravitational radiation, they will be able to detect the coalescence of IMBHs from the merger of late-type galaxies. My ongoing research into black hole mass scaling relations and BHMFs for spiral galaxies will help to establish estimates for the frequency of detections for the next generation of gravitational wave observatories. I expect this area to be of interest to prospective students and funding agencies

Other avenues for funding could focus on one or more of my main projects: (i) searching for IMBHs (§1.6), (ii) spiral galaxy evolution (§1.7), and (iii) spiral arm theory testing (§2).

1. Astronomers have discovered two sizes of black holes in the universe: SMBHs ($> 10^5\,M_\odot$) and solar mass black holes ($< 10^2\,M_\odot$). However, the so-called IMBHs theorized to fill in the evolutionary gap, remain elusive. The discovery of IMBHs would be a great boon to future research and provide potential multi-messenger targets for future generations of gravitational wave detectors. I plan to conduct an exhaustive search for IMBH candidates in the local universe ($\lesssim 10\,\mathrm{Mpc}$), where follow-up observations from ALMA, with its 20 mas resolution, could yield direct observations of the sphere-of-influence of IMBHs.

2. Legitimate high-redshift spiral galaxies at $z > 2$ constitute a brand new area of research. I would like to expand my collaboration with Tiantian to include Deanne Fisher, who is an expert on turbulent, clumpy disk galaxies, and is lead-investigator of the Dynamics of Newly Assembled Massive Objects project.[10] Tying together these local analog

[7] This software has recently been updated to run in parallel, see https://github.com/treuthardt/P2DFFT.

[8] http://sparcfire.ics.uci.edu

[9] http://www.ozgrav.org

[10] http://dynamo.swinburne.edu.au

galaxies with their high-$z$ counterparts as the predecessors to spiral galaxies could yield intriguing insights on galaxy evolution.

3. My collaboration with PHANGS is just beginning. As principle-investigator on this project, I will seek future time on ALMA and the VLT to explore spiral density waves in face-on spiral galaxies.

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